MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 07** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | $(5 \times 1 = 5$ Marks)

Q1. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45°. The height of the tower is: [1]

(A) $7(\sqrt{3}+1)$ m (B) $7(\sqrt{3}-1)$ m (C) $14\sqrt{3}$ m (D) 21 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Horizontal distance $= 7\cot 45^\circ = 7$ m. Tower height above building $= 7\tan 60^\circ = 7\sqrt{3}$ m. Total tower height $= 7 + 7\sqrt{3} = \boxed{7(\sqrt{3}+1) \text{ m}}$.

Q2. A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30°. As the car approaches the tower, the angle of depression: [1]

(A) decreases (B) increases (C) remains constant (D) becomes zero

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: For tower height h and horizontal distance d, $\tan\theta = \dfrac{h}{d}$. As the car approaches (d decreases), $\tan\theta$ increases, so the angle of depression $\boxed{\text{increases}}$.

Q3. If two towers of heights h_1 and h_2 subtend angles of 60° and 30° respectively at the mid-point of the line joining their feet, then the ratio $h_1 : h_2$ is: [1]

(A) 1 : 2 (B) 1 : 3 (C) 3 : 1 (D) $\sqrt{3}$: 1

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: Let the distance of the mid-point from each tower foot be d. Then $h_1 = d\tan 60^\circ = \sqrt{3}\,d$ and $h_2 = d\tan 30^\circ = \dfrac{d}{\sqrt{3}}$. Hence $\dfrac{h_1}{h_2} = \dfrac{\sqrt{3}\,d}{d/\sqrt{3}} = \boxed{3:1}$.

Q4. If two towers of heights x and y subtend angles of 30° and 60° respectively at the centre of the line joining their feet, then $x : y$ is equal to: [1]

(A) $1 : 3$ (B) $3 : 1$ (C) $1 : \sqrt{3}$ (D) $\sqrt{3} : 2$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Justification: With equal distance d from the centre, $x = d\tan 30^\circ = \dfrac{d}{\sqrt{3}}$ and $y = d\tan 60^\circ = \sqrt{3}\,d$.

Thus $x : y = \dfrac{1}{\sqrt{3}} : \sqrt{3} = \boxed{1 : 3}$.

Q5. **Assertion (A):** If both the height of a building and the distance of an observer from its foot are increased by 20%, the angle of elevation of the top of the building remains unchanged.

Reason (R): The trigonometric ratio $\tan\theta = \dfrac{\text{height}}{\text{distance}}$ is invariant under identical scalar scaling of numerator and denominator. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Justification: Scaling both dimensions by 1.20 gives $\tan\theta' = \dfrac{1.20\,h}{1.20\,d} = \dfrac{h}{d} = \tan\theta \implies \theta' = \theta$. Both (A) and (R) are true and (R) explains (A).

SECTION B: Very Short Answer Questions

MATHS MANTHAN | $(2 \times 2 = 4$ Marks)

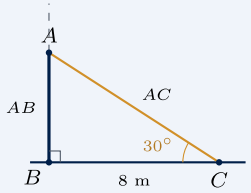
Q6. A tree breaks due to a storm and the broken part bends so that the top of the tree touches the ground making an angle of 30° with it. The distance from the foot of the tree to the point where the top touches the ground is 8 m. Find the total original height of the tree. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Let standing trunk be AB, broken bent part be AC touching ground at C ($BC = 8$ m, $\angle ACB = 30^\circ$). [0.5 Mark]

In $\triangle ABC$: $AB = 8\tan 30^\circ = \dfrac{8}{\sqrt{3}}$ m, and $AC = \dfrac{8}{\cos 30^\circ} = \dfrac{16}{\sqrt{3}}$ m. [1 Mark]

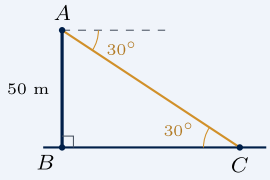
Total height $= AB + AC = \dfrac{24}{\sqrt{3}} = 8\sqrt{3} = 8(1.732) =$ $\boxed{8\sqrt{3}\text{ m} \approx 13.86\text{ m}}$. [0.5 Mark]



Q7. From the top of a cliff 50 m high, the angle of depression of a boat in the sea is 30°. Find the horizontal distance between the boat and the base of the cliff. **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Let cliff $AB = 50$ m and boat C have depression 30°, so alternate angle $\angle ACB = 30°$. [0.5 Mark]



In right $\triangle ABC$: $\tan 30° = \dfrac{50}{BC} \implies \dfrac{1}{\sqrt{3}} = \dfrac{50}{BC}$. [1 Mark]

$$BC = 50\sqrt{3} = 50(1.732) = \boxed{50\sqrt{3} \text{ m} \approx 86.60 \text{ m}}$$

Final distance: [0.5 Mark]

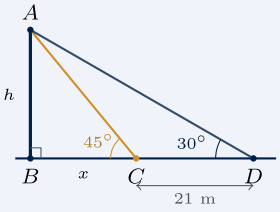
SECTION C: Short Answer Questions

MATHS MANTHAN | $(2 \times 3 = 6$ Marks)

Q8. The length of the shadow of a tower standing on level ground is found to be 21 metres longer when the sun's altitude is 30° than when it was 45°. Prove that the height of the tower is $\dfrac{21}{\sqrt{3}-1}$ m, and find its numerical value. [Take $\sqrt{3} = 1.732$] **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let tower $AB = h$ m, shadow at 45° be $BC = x$ m, and shadow at 30° be $BD = (x + 21)$ m ($CD = 21$ m). [0.5 Mark]



In right $\triangle ABC$: $\tan 45° = \dfrac{h}{x} \implies x = h$ — (i). In right $\triangle ABD$: $\tan 30° = \dfrac{h}{x+21} \implies x + 21 = h\sqrt{3}$ — (ii). [1 Mark]

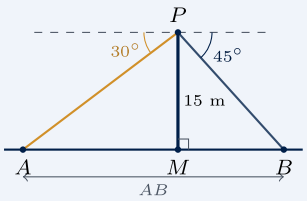
Substituting $x = h$ into (ii): $h + 21 = h\sqrt{3} \implies h(\sqrt{3} - 1) = 21 \implies h = \boxed{\dfrac{21}{\sqrt{3}-1} \text{ m}}$ [Hence Proved]. [1 Mark]

Rationalizing: $h = \dfrac{21(\sqrt{3}+1)}{3-1} = 10.5(\sqrt{3}+1) = 10.5 \times 2.732 = \boxed{28.686 \text{ m} \approx 28.69 \text{ m}}$. [0.5 Mark]

Q9. From a point on a bridge across a river, the bridge is at a height of 15 m above the water level. The angles of depression of the two opposite river banks are observed to be 30° and 45° respectively. Find the perpendicular width of the river. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let bridge point P be at height $PM = 15$ m above water surface AB, with opposite bank depressions 30° ($\angle PAM = 30°$) and 45° ($\angle PBM = 45°$). [0.5 Mark]



In right $\triangle PMA$: $AM = 15 \cot 30° = 15\sqrt{3}$ m. [1 Mark]

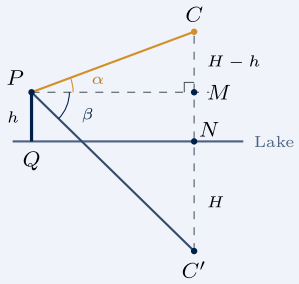
In right $\triangle PMB$: $MB = 15 \cot 45° = 15$ m. [1 Mark]

Width of the river $= AB = AM + MB = 15(\sqrt{3} + 1) = 15(1.732 + 1) = 15 \times 2.732 = \boxed{15(\sqrt{3}+1) \text{ m} \approx 40.98 \text{ m}}$. [0.5 Mark]

Q10. The angle of elevation of a cloud from a point h metres above the surface of a lake is α and the angle of depression of its reflection in the lake is β. Prove that the height of the cloud above the lake surface is: $\dfrac{h(\tan\beta + \tan\alpha)}{\tan\beta - \tan\alpha}$. [5]

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let lake surface be horizontal line QN, observation point P at height $PQ = h$ above Q. Let cloud C be at height $CN = H$ above lake, so its reflection C' lies at depth $C'N = H$ below lake. Draw $PM \perp CC'$. Then $MN = PQ = h$, $CM = H - h$, and $C'M = H + h$. [1 Mark]



Let $PM = x$. In right $\triangle PMC$: $\tan\alpha = \dfrac{H-h}{x} \implies x = \dfrac{H-h}{\tan\alpha}$ — (i). In right $\triangle PMC'$: $\tan\beta = \dfrac{H+h}{x} \implies x = \dfrac{H+h}{\tan\beta}$ — (ii). [2 Marks]

Equating (i) and (ii): $\dfrac{H-h}{\tan\alpha} = \dfrac{H+h}{\tan\beta} \implies (H-h)\tan\beta = (H+h)\tan\alpha.$

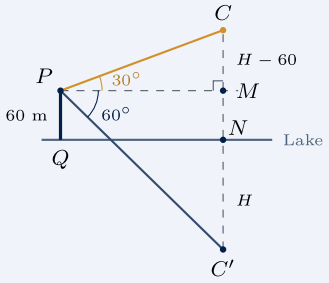
$$H(\tan\beta - \tan\alpha) = h(\tan\beta + \tan\alpha) \implies \boxed{H = \frac{h(\tan\beta + \tan\alpha)}{\tan\beta - \tan\alpha}} \quad \text{[Hence Proved]}$$

Eliminating x and solving for H: [2 Marks]

Q11. The angle of elevation of a stationary cloud from a point 60 m above the surface of a lake is 30° and the angle of depression of its reflection in the lake is 60°. Find the height of the cloud above the surface of the lake. [5]

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let $PQ = 60$ m be the height of observer P above lake QN. Let cloud C be at height $CN = H$ m above lake, so reflection C' is at depth $NC' = H$ m below lake. Here $CM = H - 60$, $C'M = H + 60$, and $PM = x$. [1 Mark]



In right $\triangle PMC$: $\tan 30° = \dfrac{H-60}{x} \implies x = \sqrt{3}(H-60)$ — (i). In right $\triangle PMC'$: $\tan 60° = \dfrac{H+60}{x} \implies x = \dfrac{H+60}{\sqrt{3}}$ — (ii). [2 Marks]

Equating (i) and (ii): $\sqrt{3}(H-60) = \dfrac{H+60}{\sqrt{3}} \implies 3(H-60) = H + 60.$

$$3H - 180 = H + 60 \implies 2H = 240 \implies \boxed{H = 120 \text{ m}}$$

Solving for cloud height H: [2 Marks]

Q12. CASE STUDY: Observation Deck at Sky Tower Architecture

An architect standing on the flat roof (P) of a commercial building of height 12 m ($PG_1 = 12$ m) observes a metropolitan Sky Tower BT located at horizontal distance 60 m ($PM = 60$ m). From P, elevation of top T is $60°$ and depression of base B is $30°$.

(i) Find the direct line-of-sight distance from the building roof (P) to the base of the Sky Tower (B). [1]

(ii) Determine the height of the section of the Sky Tower (TM) that rises above the horizontal eye-level line of the building roof. [Take $\sqrt{3} = 1.732$] [2]

(iii) Calculate the total vertical height of the Sky Tower (BT). [2]

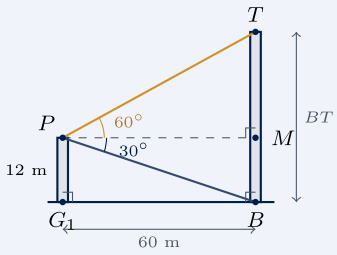
OR

If the architect looks from the base of the building (G_1) towards the top of the Sky Tower (T), write the expression for $\tan\theta$ representing the angle of elevation. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

(i) Using the given horizontal distance $PM = 60$ m and depression angle $\angle MPB = 30°$ in right $\triangle PMB$:

$$\cos 30° = \frac{PM}{PB} \implies PB = \frac{60}{\sqrt{3}/2} = \boxed{40\sqrt{3} \text{ m} \approx 69.28 \text{ m}}$$



(Note: The printed data $PG_1 = BM = 12$ m and $PM = 60$ m with $30°$ is overdetermined since $\tan 30° \neq \frac{12}{60}$; using right $\triangle PG_1B$ / PMB legs 12 and 60 gives $PB = \sqrt{12^2 + 60^2} = 12\sqrt{26} \approx 61.19$ m.) [1 Mark]

(ii) In right $\triangle PMT$ ($\angle TPM = 60°, PM = 60$ m):

$$TM = 60\tan 60° = 60\sqrt{3}$$

$$= 60(1.732) = \boxed{60\sqrt{3} \text{ m} \approx 103.92 \text{ m}}$$

Subpart (ii): [2 Marks]

(iii) Main Choice: From $\triangle PMB$ ($PM = 60$ m, $\angle MPB = 30°$), $BM = 60\tan 30° = 20\sqrt{3}$ m ≈ 34.64 m, giving trigonometric model height $BT = TM + BM = 60\sqrt{3} + 20\sqrt{3} = \boxed{80\sqrt{3} \text{ m} \approx 138.56 \text{ m}}$. (Alternatively, using printed roof height $BM = 12$ m directly gives $BT = 60\sqrt{3} + 12 = 103.92 + 12 = 115.92$ m.) [2 Marks]

(iii) OR Choice: From the ground base G_1 at horizontal distance $G_1B = 60$ m, in right $\triangle G_1BT$:

$$\boxed{\tan\theta = \frac{BT}{60}}$$

(i.e., $\frac{80\sqrt{3}}{60} = \frac{4\sqrt{3}}{3} \approx 2.309$, or $\frac{115.92}{60} = 1.932$ using $BM = 12$ m). [2 Marks]

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